

Fundamentals of ML

Spam filters · credit scoring · fraud detection · demand forecasting — none of these can be written as rules. That is the ML boundary. Everything in this week sits on it.

Before we start — is tax-bracket calculation rules or ML? Defend your answer.

WHAT'S INSIDE

- 01** Rules vs ML — the single decision
- 02** The three ML families
- 03** The 7-step supervised workflow
- 04** Metrics — what to report
- 05** Your first model in 6 lines

WHY THIS LESSON MATTERS

Rules solve problems you can articulate. ML solves the rest.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Define the three ML families with one example each.

2

Classify any business problem into the correct family in 15 s.

3

Describe the 7-step supervised workflow.

4

Compare classification vs regression metrics.

5

Train a LogisticRegression end-to-end in 6 lines.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Rules vs ML — the single...

The most valuable engineering intuition this week

02

The three ML families

Family is determined by what you HAVE, not by what algorithm you prefer.

03

The 7-step supervised...

Steps 1–4 are 80% of your impact

04

Metrics — what to report

Pick ONE primary metric up front.
Never move the goalposts.

01

PART 1 OF 4

Rules vs ML — the single...

The most valuable engineering intuition this week

CONCEPT 1 OF 4

Rules vs ML — the single decision

Rules win: few inputs, deterministic, regulation-driven.

ML wins: many inputs, fuzzy logic, labelled history exists.

Hybrid is common: rules for guardrails, ML for nuance.

Decision frame: can a domain expert write < 20 rules? If yes, rules.

HEURISTIC

< 20 rules?

Yes → rules. No → ML. Simple. Usable today.

DEFINITION

Rules vs ML — the single decision

The most valuable engineering intuition this week

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

The three ML families

Family is determined by what you HAVE, not by what algorithm you prefer.

KEY POINTS

- Family is determined by what you HAVE, not by what algorithm you prefer
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

The 7-step supervised workflow

1. Define the question + cost per error.
2. Collect and label data.
3. Engineer features.
4. Split train / val / test.
5. Train. 6. Evaluate. 7. Deploy + monitor.

80%

Steps 1-4

Shoulders-down on data quality before you tune any hyperparameter.

DEFINITION

The 7-step supervised workflow

Steps 1-4 are 80% of your impact

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Metrics — what to report

Pick ONE primary metric up front. Never move the goalposts.

KEY POINTS

- Pick ONE primary metric up front. Never move the goalposts
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Classify these 5 Indian systems

PROMPT

Assign each: UPI fraud • IRCTC Tatkal • Swiggy ETA • Aadhaar face-match • Ola surge pricing.

HOW TO RUN IT

- 1 Assign each: UPI fraud • IRCTC Tatkal • Swiggy ETA • Aadhaar face-match • Ola surge...
- 2 Defend with one sentence per system.
- 3 Room aligns on disagreements.
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which of the three families fits the first project you want to build?

Q2

Where have you been seduced by "let's try ML" when rules would do?

Q3

Which metric will you default to for your first classification problem?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Predicting whether a loan defaults from 50 features is:

A

Supervised classification

B

Unsupervised clustering

C

RL

D

Regression

Predicting whether a loan defaults from 50 features is:



CORRECT ANSWER

Supervised classification

WHY

Labels exist (default Y/N). It is supervised classification

Accuracy misleads most when:

A

Features are many

B

Classes are imbalanced

C

Model is linear

D

Data is tidy

Accuracy misleads most when:

CORRECT ANSWER



Classes are imbalanced

WHY

Predicting the majority class always yields high accuracy on imbalanced data. Use F1 / ROC-AUC

Why NEVER let test data touch the fit?

A

Speed

B

Data leakage

C

Memory

D

Compliance

Why NEVER let test data touch the fit?

CORRECT ANSWER



Data leakage

WHY

Leakage inflates the apparent score and hides overfitting

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

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THE TAKEAWAY

"Rules solve problems a human can articulate. ML solves the rest."

WHAT TO NOTICE

1

Rules vs ML — the single...

2

The three ML families

3

The 7-step supervised...

4

Metrics — what to report

Mini assignment — train end-to-end

On any dataset you own, train a LogisticRegression or LinearRegression end-to-end in 20 lines.

DELIVERABLES

- State the question in one sentence.
- Split train / test 80 / 20.
- Train, score, pick ONE primary metric.
- Report that metric + one honest limitation.
- Push notebook to GitHub.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Notebook URL + 2-sentence write-up on the forum.

ONE THING TO REMEMBER

“

"Rules solve problems a human can articulate. ML solves the rest."

— AI Learning Hub

END OF LESSON 3

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 2 — Neural Networks. The perceptron, XOR, and your first MLP.